

Name of Student: Prajwal Agrawal

PRN: 25070122136

Semester: III

Year: AY 2026-27

Subject: Programming with Java

ASSIGNMENT No: 1

TITLE: Java Environment Setup and Basic Program Execution

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programme

```
J ConstructorDemo.java
1  class Student {
2      String name;
3      int rollNo;
4
5      // Default Constructor
6      Student() {
7          name = "Unknown";
8          rollNo = 0;
9      }
10
11     // Parameterized Constructor
12     Student(String name, int rollNo) {
13         this.name = name;
14         this.rollNo = rollNo;
15     }
16
17     void display() {
18         System.out.println("Student Name: " + name);
19         System.out.println("Roll Number : " + rollNo);
20     }
21
22     Run | Debug
23     public static void main(String[] args) {
24         Student s1 = new Student();
25         Student s2 = new Student(name: "Prajwal", rollNo: 101);
26
27         System.out.println(x: "Default Constructor:");
28         s1.display();
29
30         System.out.println(x: "\nParameterized Constructor:");
31         s2.display();
32     }
}
```

Output

```
Default Constructor:  
Student Name: Unknown  
Roll Number : 0  
  
Parameterized Constructor:  
Student Name: Prajwal  
Roll Number : 101
```

Conclusion

Questions

1. Explain the main concept used in Java environment setup, variables and methods.
2. What are the advantages of using Java variables and methods?
3. How can this implementation be improved?
4. What are the real-world applications?